Shashank Kirtania

Doctorate Student
University of Michigan, Ann Arbor

shashankkirtania123@gmail.com
5hv5hvnk.github.io | GitHub |
LinkedIn | Google Scholar

Education

University of Michigan, Ann Arbor — 2026 - Present
Ph.D. in Computer Science — *Advised by: Prof. Gabriel Poesia*

Thapar Institute of Engineering and Technology — 2019 - 2023
BTech. in Computer Science

Experience

Microsoft Research, India — Sep. 2025 – May 2026
Research Fellow (Predoctoral) — *Bengaluru*
Projects: *Verified Code Generation*

Microsoft, India — Sep. 2023 – Sep. 2025
Research Fellow (Predoctoral) with the PROSE group — *Remote / Bengaluru*
Projects: StackFeed, MetaReflection, Logic-LM++

Wadhwani AI — Jan. 2023 - July 2023
Machine Learning Research Intern — *Remote / Bengaluru*
Project: Oral Reading Fluency for Indic languages

Google Summer of Code, PyMC — May 2022 - Sep 2022
Open Source Contributor — *Remote*
Project: APIs for probabilistic ML models.

Publications And Preprints

C = Conference, W = Workshop, J = Journal, P = Preprint

P.1 ***"Code is cheap, show me the specification": Training LLMs for end-to-end Verified Code Generation***
Swayam Singh*, Ramneet Singh*, **Shashank Kirtania***, Tianyu Cheng, Aditya Kanade, Shan Lu, Nagarajan Natarajan
Under review — **[2026]**

C.1 ***Improving Language Agents through BREW: Bootstrapping expeRientially-learned Environmental knoWledge***
Shashank Kirtania, Param Biyani, Priyanshu Gupta, Yasharth Bajpai, Roshni Iyer, Sumit Gulwani, Gustavo Soares
Conference on Language Modeling 2026 — **[COLM'26]**
Also presented at Multi-Turn Interactions in LLMs at NeurIPS 2025 — **[MTI @ NeurIPS'25]**

W.1 ***IndiMathBench: Autoformalizing Mathematical Reasoning Problems with a Human Touch***
Param Biyani, **Shashank Kirtania**, Yasharth Bajpai, Sumit Gulwani, Ashish Tiwari
Formal Methods in AI Workshop at AAAI 2026 — **[FMAI @ AAAI'26]**

C.2 ***STACKFEED: Structured Textual Actor-Critic Knowledge Base Editing with Feedback***
Shashank Kirtania*, Naman Gupta*, Priyanshu Gupta, Krishna Kariya, Sumit Gulwani, Arun Iyer, Suresh Parthasarathy, Arjun Radhakrishna, Sriram K. Rajamani, Gustavo Soares
Empirical Methods in Natural Language Processing 2025 — **[EMNLP'25]**

W.2 ***Activation Steering in Neural Theorem Provers***
Shashank Kirtania, Arun Iyer
Formal Methods in the Wild Workshop at ICLR 2025 — **[FMWild @ ICLR'25]**

C.3 ***MetaReflection: Learning Instructions for Language Agents using Past Self-Reflections***
Priyanshu Gupta*, **Shashank Kirtania***, Ananya Singha*, Sumit Gulwani, Arjun Radhakrishna, Sherry Shi, Gustavo Soares
Empirical Methods in Natural Language Processing 2024 — **[EMNLP'24]**

W.3 ***Logic-LM++: Multi-Step Refinement for Symbolic Formulations***
Shashank Kirtania, Priyanshu Gupta, Arjun Radhakrishna
Natural Language Reasoning Workshop at ACL 2024 **[NLRSE @ ACL'24]**

J.1 ***DWT-CompCNN: Deep Image Classification Network for High-Throughput JPEG2000 Compressed Documents***
Tejasvee Bisen, M Javed, **Shashank Kirtania**, P Nagabhushan
Pattern Analysis and Applications **[Springer 2023]**

Selected Projects And Collaborations

Automated Theorem Proving — Dec. 2024 - Present
Advisor: Dr. Arun Iyer, Dr. Emiliy First, Dr. Ashish Tiwari — *MSR India*

- **Identified transformer circuits** in foundational models to create activation patches for manipulating formal proof writing. Presented at FM in Wild @ ICLR 2025
- Worked on **formalizing 400+ problems for INMOBench** in lean, and did a empirical study of LLMs and their failures in autoformalization. Oral Presentation at FM for AI workshop @ AAAI
- Working on **Mechanistic interpretability** of LLMs to improve their **ability of mathematical reasoning**.

Automated Code Repair — Jan. 2024 – June 2024
Advisor: Dr. Gustavo Soares, Dr. Arjun Radhakrishna — *Microsoft PROSE*

- Developed **MetaReflection**, an offline reinforcement learning technique **enhancing language agents by leveraging past failures**, improving performance in code **vulnerability detection**. Published @ EMNLP Main.
- This technique showed significant improvements in low resource programming languages and in distilling capabilities of LLMs into SLMs, It is currently **used by GitHub for secret key detection**. deployed by Github Enterprise in 2024
- Worked on **fixing component governance issues from alerts for C#** using tool-aided LLM system. Developed Roslyn APIs as tools to improve code understanding for LLMs.

Knowledge Learning from LLM Interactions — Aug. 2024 – Present
Advisor: Dr. Arun Iyer, Dr. Gustavo Soares — *MSR India*

- Developed **STACKFEED**, a multi-agent actor-critic framework for **iterative knowledge base refinement** using expert feedback, enhancing Retrieval-Augmented Generation (RAG) systems by up to 8% over baselines.
- Built pipeline to use feedback to edits in knowledge bases for **code migration and generation.** Published @ EMNLP 2025
- Worked on learning knowledge bases from interactions of LLM agents Published at MTI @ NeurIPS

Language Models for Automated Reasoning — Apr. 2024 – Present
Advisor: Dr. Arun Iyer, Dr. Arjun Radhakrishna

- Introduced **Logic-LM++**, a **multi-step refinement framework** leveraging LLMs for symbolic reasoning, achieving significant improvements over existing methods. Published at NLRSE Workshop @ ACL
- Built pipeline to capture **activation vectors** representing natural language reasoning for LLMs. Used this technique to imporve performance on *miniF2F-test* and *PutnamBench*. Published at FM in Wild Workshop @ ICLR.

Automated Repository Code Migration — Sep. 2023 – May 2024
Advisor: Dr. Gustavo Soares, Dr. Sumit Gulwani — *Microsoft PROSE*

- Built pipeline for using LLMs as pseudo-domain experts that learn from failures to detect code vulnerability.
- Employed novel strategy of automatic instruction learning to optimize performance of LLM agent trajectories.

Achievements and Position of Responsibilities

- Among top 1,200 students selected for **Google Summer of Code 2022** with Numfocus (PyMC).
- Reviewer at AI4Math workshop at ICML 2025 and at DL4Code workshop at NeurIPS 2025 and at ICML 2026.
- Volunteer at PLDI 2024, ICLR 2025.
- Positioned **6th nationally among 5000 + teams** in Amazon Machine Learning challenge 2023.
- **Bronze Medal** in American Express Time Series Forecasting Challenge 2022.
- PyData Delhi talk on "Introduction to Probabilistic Programming".